KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE SCIENCE
DEPARTMENT OF PHYSIOTHERAPY

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN PHYSIOTHERAPY

PAPER: THERAPEUTIC SCIENCES

DATE: 22ND JANUARY 2025	**TIME: 3 HOURS (9:00PM – 12:00 PM)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER ..01/PHS/013/AM-1CB 25/122/079.....

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SECTION SHORT ANSWER QUESTIONS 40 MARKS

1. Exercise is an essential part of anterior shoulder dislocation rehabilitation program as it can have a positive effect on muscular tissue and muscular strength. A physical therapist gradually initiates a group of exercises that involve constant velocity of shoulder movement with variable resistance. The resistance is given throughout the muscle action to allow maximum tension. This type of exercise is classified as:
   A. Isokinetic
   B. Isometric
   C. Isotonic
   D. Endurance exercise
2. A physical therapist is observing a patient who is walking with crutches. The patient advances both crutches forward, and then swing both legs past the crutches at the same time. Based on the sequence, the patient is demonstrating which crutch gait?
   A. Two-point crutch gait
   B. Swing-to crutch gait
   C. Swing-through crutch gait
   D. Tripod crutch gait
3. Range-of-motion exercises are exercises that move a joint through the degree of its limitations. Which type of range-of-motion exercises involves movement of a joint done by the physical therapist or a mechanical device without any muscle contraction exerted by the patient?
   A. Active-assistive range of motion exercise
   B. Active range of motion exercise
   C. Passive range of motion exercise
   D. Stretching exercise
4. Open-chain exercises or closed-chain exercises are usually part of the exercise prescription given to a patient who is referred for exercise therapy. Which of the following activities is classified as an open chain exercise?
   A. Bicep curls
   B. Squats
   C. Push-ups
   D. Leg press
5. You are carrying out manual muscle testing of the left infraspinatus muscle of a patient whom you suspect has rotator cuff insufficiency. Which of the following positions best isolates this muscle for testing of its strength?
   A. Arm is placed to the side; resist external rotation.
   B. Arm is placed in neutral position and with the thumb up; resist abduction.
   C. Arm is maximally internally rotated up the back with the back of the hand against the back; lift the arm off the back.
   D. Arm is placed up the back, at least the sacrum level, with the back of the hand facing the patient's arm, and the elbow is flexed to 90-degrees; arm is held at that position.

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6. A male patient complains of pain in the inner aspect of the left knee. Assessment findings include swelling and knee instability when making a cut. An injury involving the medial collateral ligament is suspected. On valgus stress test, a laxity of 7 mm is noted. Which of the following interventions is least likely implemented to the patient?
   A. Bracing with a short-hinged knee orthosis that blocks 20-degrees of terminal extension but allows full flexion.
   B. Initial non-weight bearing on the affected extremity.
   C. Crutches for the first few days.
   D. Closed-chain exercises
7. A physical therapist uses a type of massage that involves striking of the soft tissue with repetitive blows, using the hands in a rhythmic and rapid manner. The therapeutic effect of this massage is caused by the compression of trapped air on impact. This type of massage is:
   A. Deep friction
   B. Effleurage
   C. Pétrissage
   D. Tapotement
8. Pétrissage involves compression of skin and muscle between the fingers and thumb. Which of the following treatment approaches mainly focuses on the execution of diagonal, spiraled patterned movements to improve muscle strength and enhance neuromuscular control?
   A. Neurodevelopmental treatment
   B. Proprioceptive neuromuscular facilitation
   C. Brunnstrom's movement
   D. Muscle re-education approach
9. A patient with a significant limitation in range of motion of the left shoulder is referred to the clinic for rehabilitation. The physical therapist uses a technique that passively moves the affected extremity through an antagonistic pattern until resistance is felt. At this point, the patient is told to isotonically contract the antagonistic muscles in a diagonal pattern as the therapist applies resistance. After this contraction, the patient is told to relax, and after interval the therapist repeats the technique from the start. The proprioceptive neuromuscular facilitation technique used is:
   A. Repeated contraction
   B. Hold-relax contract
   C. Hold-relax
   D. Contract-relax
10. A male patient is referred to physical therapy for the treatment of anterior glenohumeral instability. During the acute phase, the physical therapist is correct to include which of the following interventions?
   A. Active range-of-motion exercises concentrating on the joint capsule
   B. Pendelum exercises
   C. High-speed, high-energy strengthening exercises
   D. Isotonic dumbbell resistance exercises with the supraspinatus, internal rotators, and external rotators
11. A physical therapist uses a proprioceptive neuromuscular facilitation technique of repeated contraction. This specific technique uses two basic patterns for the shoulder and hip. For the shoulder motions, D1 is referred to as:
   A. Flexion-abduction-external rotation
   B. Flexion-adduction-external rotation
   C. Extension adduction-internal rotation
   D. Extension-abduction-internal rotation

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12. A physical therapist is planning the treatment program for a patient with cervical radiculopathy. Which of the following is the most appropriate intervention during the recovery stage of the disease?
    A. Manual traction
    B. Maintenance of proper head and neck positioning in everyday activity
    C. Initiation of resistive cervical stabilization exercises in various planes before isometric strengthening
    D. Strengthening of the scapular stabilizing muscles.
13. A male elderly with osteoarthritis of the right knee is ordered to use a single-point cane to improve balance when walking. A physical therapist provides the following instructions when ambulating with a single-point cane on a level surface except:
    A. The affected leg and the cane should swing and strike the ground at the same time.
    B. Weight is shifted through the arm holding the cane, as needed.
    C. Hold the cane in the hand on the affected side.
    D. Ensure that the unaffected leg fully bears weight on the first step on level surfaces.
14. When taking goniometric measurements of medial rotation of the shoulder, where should the moveable arm of the goniometer be aligned?
    A. Parallel with the humerus toward the coracoid process.
    B. Parallel with the humerus toward the acromion process.
    C. Parallel to the ulna toward the ulnar styloid process.
    D. Parallel to the radius toward the 2nd metacarpal.
15. A patient with type IV SLAP lesion had a surgery that involved resection of the affected area of the tendon. After the surgery, the patient is immediately referred for therapy. The physical therapist includes which of the following interventions?
    A. Active-assisted forward flexion with the use of pulleys
    B. Active-assisted external rotation movements
    C. Wrist and elbow active range of motion exercises
    D. Endurance training with the use of isokinetic exercises
16. A patient is receiving physical therapy treatment for weakness and difficulty transferring following an above-knee amputation approximately 4 weeks ago. Which of the following will be the MOST important initial treatment to improve gait mechanics?
    A. Have the stump fitted for a custom prosthesis
    B. Begin performing progressive side lying hip abduction exercises
    C. Perform prone hip extension stretching
    D. Initiate progressive well-leg strengthening
17. A patient with type II diabetes is coming to the physical therapy clinic for treatment after a total knee replacement. Which of the following will be most important when treating this individual with therapeutic exercise?
    A. Include activities for large muscle groups to improve glucose absorption.
    B. Encourage the patient to eat more than ½ hour before the therapy session.
    C. Add appropriate exercises as tolerated, focusing on knee extension.
    D. Instruct the individual to bring small snacks to eat throughout the session.

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18. A patient is having difficulty maintaining his balance during static standing, especially in crowded environments. Which of the following balance strategies is the most active during this activity?
    A. Ankle strategy
    B. Hip strategy
    C. Suspensory strategy
    D. Stepping strategy
19. When using the parallel bars for gait training, what is the appropriate adjustment of height that will optimize stability and security for the patient?
    A. A height that allows for elbow flexion of approximately 0 degrees.
    B. A height that allows for elbow flexion of approximately 10 degrees.
    C. A height that allows for elbow flexion of approximately 15 degrees.
    D. A height that allows for elbow flexion of approximately 20 degrees.
20. A patient receiving physical therapy for a recent shoulder injury demonstrates a 2+/5 manual muscle test for shoulder flexion. Which of the following BEST describes the amount of motion this patient can perform?
    A. The patient is able to complete more than ½ of the available range of motion against gravity.
    B. The patient is able to complete less than ½ of the available range of motion against gravity.
    C. The patient is able to complete the available range of motion with gravity eliminated.
    D. The patient is able to complete less than ½ of the available range of motion with gravity eliminated.
21. A 32-year-old female presents to the clinic complaining of left shoulder pain. The patient's symptoms include pain with reaching and limited motion in all planes. The patient's symptoms have been progressively worsening over the last month. What would be the MOST effective initial treatment?
    A. Refer the patient to their primary care provider for intra capsular corticosteroid injections.
    B. Begin gentle progressive stretching exercises with the focus on increased active range of motion.
    C. Initiate rotator cuff strengthening exercises, focusing on external rotation.
    D. Instruct the patient on Codman pendulum exercises and apply a moist heat pack.
22. Transcutaneous electrical nerve stimulation is contraindicated to patients with a demand-type of pacemaker and to women who are pregnant. In addition to these, the TENS should also be placed in the right areas to prevent injuries and complications. TENS should not be placed in the following areas except:
    A. Carotid sinuses
    B. Anterior neck
    C. Area of sensory impairment
    D. On the skin over the painful area
23. A female patient with muscle contractures is being considered for microwave diathermy, a deep heat modality. As the physical therapist reviews the patient's history, which of the following details of information contraindicates the procedure?
    A. A recent cardiac pacemaker procedure
    B. A diagnosis of periostitis
    C. A history of rheumatoid arthritis
    D. A diagnosis of bursitis

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24. Cryotherapy, also known as cold therapy, makes use of ice or cold applications for the treatment of certain musculoskeletal disorders or injuries. The devices and substances utilized cause the withdrawal of heat. This modality is contraindicated to patients with the following conditions except:
    A. Hypertension
    B. Raynaud's disease
    C. Upper motor neuron spasticity
    D. Rheumatoid arthritis
25. A female patient with musculoskeletal pain is referred to the clinic for physical therapy. The therapist plans to include transcutaneous electrical nerve stimulation application as one of the modalities for the patient. Which of the following details of information in the patient record makes TENS a contraindicated modality to the patient?
    A. A diagnosis of diabetes neuropathy.
    B. A diagnosis of spinal cord injury.
    C. A history of angina pectoris.
    D. Use of a demand-type of pacemaker.
26. A physical therapist chooses to use therapeutic ultrasound for deep heating of the quadriceps muscle. Which set of ultrasound parameters will be MOST effective at increasing the temperature of tissue 3.0 centimeters deep?
    A. 1.0 MHz, 1.5 W/cm², 50% duty cycle
    B. 3.0 MHz, 1.5 W/cm², 100% duty cycle
    C. 1.0 MHz, 1.5 W/cm², 100% duty cycle
    D. 3.0 MHz, 1.5 W/cm², 50% duty cycle
27. A patient is being treated for flexor hallucis longus tendonitis. Which of the following modalities will be MOST useful in decreasing pain and speeding recovery?
    A. Topical menthol
    B. Iontophoresis
    C. Transcutaneous electrical nerve stimulation
    D. Ultrasound
28. Which of the following treatment parameters would be MOST appropriate for a patient receiving intermittent lumbar traction for a diagnosis of multifidus muscle spasms?
    A. 30 minutes, 10% body weight traction
    B. 30 minutes, 25% body weight traction force
    C. 30 minutes, 50% body weight traction force
    D. 30 minutes, 75% body weight traction force
29. The modalities of deep heating convert energy into heat as it goes through body tissues. Electromagnetic radiation, high frequency sound and high-frequency currents are some of the energy sources used. Which of the following heating modalities is considered as the best method for large-area deep heating?
    A. Microwave diathermy
    B. Hydrotherapy
    C. Ultrasonography
    D. Shortwave diathermy

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30. A patient with adhesive capsulitis is referred to the clinic for physical therapy. In combination with a physical therapy program that consists of range of motion and stretching exercises, deep heat is one of the modalities to improve shoulder mobility. Which of the following modalities is the most ideal for the patient?
    A. Shortwave diathermy
    B. Microwave diathermy
    C. Electric heating pad
    D. Ultrasound
31. A normal subject raises the right upper extremity from a position of 60 degrees of shoulder flexion to 120 degrees of shoulder flexion. What type of muscular contraction is occuring in anterior deltoid during this activity?
    A. Concentric
    B. Eccentric
    C. Isometric
    D. deltoid is not active
32. Movements during ambulation are the result of several factors, including those produced by the muscles. Muscles are essential for acceleration, deceleration, and shock absorption. Which of the following is true about the muscles involved with gait kinetics?
    A. Acceleration is the result of the muscle's eccentric activity.
    B. The greatest amount of muscle activity is required during the first 10% of the stance phase and last 10% of the swing phase.
    C. The times of least muscle activity are during periods of acceleration and deceleration.
    D. Locomotion requires continuous muscle activity as it involves movements from one location to another.
33. A physical therapist conducts a gait assessment and balance evaluation test on a patient. Which of the following findings obtained during the test demonstrates an abnormal gait pattern, requiring referral for further evaluation?
    A. At foot contact, the dorsiflexors eccentrically contract and the foot plantar flexes from heel strike to a foot-flat position.
    B. The hip joint moves from 20 to 30-degrees of flexion at initial contact to 10-degrees of extension in terminal stance as the trunk smoothly moves over the stance limb.
    C. At midstance, the pelvis shifts toward the swing leg; lateral movement of the hip toward the stance leg is observed.
    D. On initial contact and heel strike, the knee is fully extended.
34. The ankle and foot work together during gait to provide the balance needed as the body moves over an uneven ground. The following indicate the normal averages and expected findings from mid-stance to toe-off except:
    A. The interphalangeal joints are in neutral position.
    B. The ankle moves from neutral to 10-degrees of dorsiflexion, progressing to
    C. about 20-degrees of plantar flexion on toe-off.
    D. The longitudinal arch lengthens.

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35. Gait assessment and balance evaluation is conducted on a patient who recently resumed full weight bearing after an injury. On assessment, a drop of the pelvis is noted toward the swing leg with visible movement of the hip toward the stance leg. The patient moves the trunk laterally over the stance-phase leg. The patient's gait indicates:
    A. Quadriceps weakness
    B. Hip abductor weakness
    C. Hip extensor weakness
    D. Hip flexion contracture
36. On gait assessment and balance evaluation, the patient keeps the knee extended at heel strike and through the stance phase. Knee flexion is reduced. The movement into full knee extension snaps the affected knee back. This finding indicates that the patient has:
    A. Ankle dorsiflexion weakness
    B. Quadriceps weakness
    C. Hip extensor weakness
    D. Knee flexion contracture
37. While examining a patient, you notice that he has hypermobility of the talocrural joint of his left foot which is limiting dorsiflexion of the foot. Which of the following gait patterns is most likely to be observed due to this impairment?
    A. Steppage gait
    B. Double step gait
    C. Trendelenburg gait
    D. Circumduction gait
38. On gait assessment and balance evaluation, the patient demonstrates difficulty in advancing the trunk forward in terminal stance. The pelvis flexes forward to maintain vertical orientation of the trunk. Increased lumbar spine extension is observed. Based on these findings and on the gait pattern of the patient, which of the following conditions is most likely?
    A. Hip extensor weakness
    B. Hip abductor weakness
    C. Hip pain
    D. Hip flexion contracture
39. Logically, the force vector which represents the pull of a particular muscle would have its point of application located at the:
    A. Origin of the muscle
    B. insertion of the muscle
    C. axis of motion of the joint it affects
    D. intersection of its action line and moment arm
40. In gait, at preferred rate of ambulation, during transition from the static to dynamic base of support:
    A. the base narrows and foot toes-out
    B. the base widens and foot toes-out
    C. the base narrows and foot toes-in
    D. the base widens and foot remains in neutral position

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SECTION 2 SHORT ANSWER QUESTIONS 40 MARKS

41. Briefly explain:
    a) The function of postural drainage? (2 Marks)
    b) When is postural drainage done? (3 Marks)
42. Briefly explain:
    a) What is the function of chest percussion? (2 Marks)
    b) What is the purpose of chest percussion? (3 Marks)
43. Briefly explain about spirometry. (5 marks)
44. Briefly explain how you would carry out manual muscle testing for the deltoid muscle. (5 marks)
45. Define contrast bath and describe its therapeutic uses. (5 marks)
    i. Definition: (2 marks)
    ii. Therapeutic benefits: (3 marks)
46. Explain production of Interferential therapy. (5 marks)
47. Define the following terminologies: (5 marks)
    i. Kinematics:
    ii. Kinetics:
48. The stance phase occupies 60% of the total gait cycle and is divided into five sub-phases which are: (5 marks)

SECTION 3 LONG ANSWER QUESTION 20 MARKS

49. Discuss about
    a. Concentric Contractions: (10 marks)
    b. Eccentric Contractions: (10 marks)

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